

SAICHITRA NAGARAGERI

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PROFESSIONAL SUMMARY

Senior Data & AI/ML Engineer with 7+ years building scalable data platforms, ML feature and inference pipelines, and GenAI infrastructure across GCP, AWS, and Azure. Specializes in the data and ML infrastructure that powers machine learning and generative-AI applications from BigQuery and Spark pipelines to Vertex AI feature stores, embedding/RAG pipelines, and MLOps. Strong software-engineering fundamentals (Python OOD, REST APIs, microservices, CI/CD, infrastructure-as-code) and a track record of bridging engineering, data-science, and business teams to ship production systems at petabyte scale.

TECHNICAL SKILLS

AI / ML & GenAI: Vertex AI, Gemini, LLMs, RAG, embeddings & vector search (Vertex Vector Search, pgvector), LangChain, prompt engineering, Spark MLLib, scikit-learn, feature engineering, MLOps (Vertex AI Pipelines, MLflow), model serving & monitoring, SAS, R

Big Data & Distributed Systems: Apache Spark (PySpark, Spark SQL, Scala), Spark Streaming, Hadoop, Hive, Kafka, NiFi, Sqoop, HBase, Impala, Apache Beam

Cloud GCP: BigQuery, Dataflow, Dataproc, Pub/Sub, Cloud Storage, Cloud Functions, Cloud Scheduler, Vertex AI, Kubernetes Config Connector

Cloud AWS: EMR, Glue, Redshift, S3, EC2, RDS, EBS, Auto Scaling, Elastic Load Balancer

Cloud Azure: Data Factory (ADF), Databricks, Synapse Analytics, ADLS, Cosmos DB, HDInsight

Software Engineering: Python (OOP/OOD), REST APIs (FastAPI/Flask), microservices, web services, distributed systems, Git/GitLab, CI/CD (Jenkins, Maven), infrastructure-as-code, unit/integration testing, code review

ETL / ELT & Orchestration: Apache Airflow, Azure Data Factory, AWS Glue, Informatica, Talend, IBM DataStage, SSIS, Oozie

Databases & Warehousing: BigQuery, Snowflake, Redshift, PostgreSQL, Oracle, SQL Server, Teradata, Netezza, MongoDB, Cosmos DB

Data Analysis & Visualization: Advanced SQL, Python (Pandas, NumPy, SciPy), Tableau, Power BI, QlikView, EDA, statistical modeling

Data Modeling & BI: Dimensional modeling, fact/dimension tables, SCD, CDC, data warehousing, data quality, metadata management, OLAP/OLTP, file formats (Avro, ORC, Parquet)

Languages: Python, Scala, SQL, PL/SQL, T-SQL, Unix Shell Scripting

PROFESSIONAL EXPERIENCE

Senior Data Engineer AI/ML & GenAI Infrastructure

Feb 2024 – Present

Sam's Club

Dallas, TX

- Led end-to-end migration of enterprise-scale pipelines to Google Cloud Platform (BigQuery, GCS, Cloud Dataflow), consolidating petabyte-scale workloads to modernize processing, improve pipeline reliability, and reduce operational overhead.
- Built embedding-generation and vector-indexing pipelines using Vertex AI embeddings and vector search (Vertex Vector Search / pgvector), powering RAG-based semantic search over enterprise data catalogs and cutting data-discovery time for analytics teams.
- Developed an LLM-powered text-to-SQL and metadata-documentation layer on Gemini (Vertex AI) that auto-generates BigQuery query suggestions and table descriptions, reducing analyst query-authoring effort and onboarding time.
- Engineered reusable feature pipelines feeding ML models in Vertex AI, automating training-data preparation, validation, and scheduled refresh from BigQuery to enable reproducible model training and faster experimentation.
- Operationalized model serving and monitoring with Vertex AI Pipelines integrated into existing Dataflow and Pub/Sub ingestion, establishing MLOps practices for deployment, versioning, and drift detection.

- Designed a cloud-native data-completeness and quality-validation framework across 20+ production pipelines, adding LLM-based semantic anomaly detection to flag inconsistent records beyond rule-based checks and reducing data incidents.
- Tuned BigQuery performance through advanced partitioning, clustering, and execution-plan analysis, cutting query costs and improving SLA compliance for downstream reporting; tuned Apache NiFi workflows for higher throughput and fault tolerance.
- Provisioned Cloud Functions, Cloud Schedulers, and Pub/Sub topics as infrastructure-as-code via Kubernetes Config Connector (KCC), and integrated Snowflake with Dataflow, Dataproc, and BigQuery for seamless end-to-end pipelines.

Data Engineer ML Pipelines & Big Data

Apr 2022 – Jan 2024

PayPal

Austin, TX

- Built and optimized large-scale ingestion and transformation frameworks using PySpark, Spark SQL, Hive, Kafka, and Hadoop on Amazon EMR, processing data from Salesforce, SQL Server, Oracle, and SAP.
- Engineered feature-engineering pipelines in PySpark to prepare training datasets for risk and forecasting models, applying partitioning, memory tuning, and parallelism to efficiently process large-scale data.
- Developed batch and near-real-time inference pipelines with Spark Streaming, scoring data from S3 and messaging systems and persisting curated predictions to Amazon Redshift and HDFS.
- Designed, developed, and tested ETL workflows in AWS Glue, migrating campaign and transactional datasets (S3, ORC, Parquet, flat files) into Redshift for enterprise analytics and reporting.
- Architected scalable, fault-tolerant cloud infrastructure with EC2, S3, RDS, EBS, Auto Scaling Groups, and Elastic Load Balancers, enabling high-availability and elastic data-processing workloads.
- Partnered with data-science teams to productionize reproducible training-data snapshots and feature sets, and built Tableau dashboards for operational monitoring and business insights; validated MapReduce/Pig/Hive transformations against source systems for data correctness.

Senior Data Engineer Data Platforms & Predictive Modeling

May 2019 – May 2021

DXC Technology

India

- Architected and led end-to-end platform migration from Google BigQuery to Salesforce across North America and EU, designing transformation pipelines, schema-reconciliation logic, and validation frameworks that ensured zero data loss.
- Built and validated predictive models (regression, decision trees, clustering, text mining) using SAS and R on high-volume, high-granularity big-data environments, translating analytical findings into actionable business recommendations.
- Designed large-scale analytics pipelines on BigQuery and Pub/Sub modeling supplier performance and implemented PostgreSQL-backed APIs for inventory forecasting, improving forecast accuracy by 25% across hundreds of suppliers.
- Enforced Python coding standards, OOD patterns, and PR-based code-review workflows; owned CI/CD pipelines, build/release processes, and production operations on AWS (S3, EC2, RDS/PostgreSQL).
- Implemented batch vs. real-time inference pipelines with measurable latency/throughput optimization, applying web-services and distributed-systems design judgment across multi-team engagements and design reviews.

EDUCATION

Master of Science, Data Science

Dec 2022

Saint Peter's University, Jersey City, NJ